

Exercise Sheet 4

Thursday, January 15, 2026

4.1 Earliest Deadline First (EDF) Implementation

In this exercise, you will implement the EDF scheduling algorithm in FreeRTOS. By default, FreeRTOS uses a fixed-priority scheduling policy. To implement an EDF scheduler you need to modify the FreeRTOS kernel.

Hint: Your debugging tool will be of great help to observe the scheduling behavior of your implementation.

- a. Implement the EDF algorithm in FreeRTOS. You need to modify (at least) the following:
 - Task Control Block (TCB) to realize periodic tasks
Hint: Add period, worst-case execution time and relative deadline parameters
 - Scheduling policy, i.e., the mechanism which selects the next task to run
- b. Realize a workload, where you can clearly observe the effects of EDF scheduling. You can use your debugging tool to trace the job releases and job executions.
- c. Measure the overhead of your EDF implementation and compare it to the original FreeRTOS FP implementation. For this, create a workload with a high number of tasks and short periods, however, pay attention that the workload is still feasible. You can check for deadline misses by observing whether a job is still running when the next job of the same task is released, e.g., this can be realized directly in the `vTaskDelayUntil` function.
- d. Optimize your EDF implementation to reduce the overhead. **Hint:** The choice and implementation of the data structures that hold tasks has a great impact on the overhead.